

One-Step Equations

AQA · KS3 Year 8

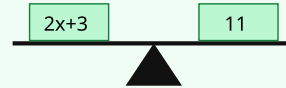
Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Only changing one side of the equation.

Wrong inverse: the inverse of $\times 3$ is $\div 3$, not -3 .

For $x \div 4 = 3$, multiply both sides by 4.

Sign slips: $x + 7 = 2$ gives $x = -5$.

METHOD

- 1 See what was done to the letter
- 2 Choose the inverse operation
- 3 Do it to BOTH sides
- 4 Simplify to leave the letter alone
- 5 Check by substituting

WORKED EXAMPLE

Solve $x + 7 = 12$.

Inverse of $+ 7$ is $- 7$: $x = 12 - 7 = 5$.

Check: $5 + 7 = 12$

VISUAL AID

Balance scale: $x + 7$ on the left pan equals 12 on the right pan. Remove 7 from both pans so the scale stays level, leaving $x = 5$.

1. (2 marks)

Solve: a) $x + 5 = 12$ b) $y - 3 = 10$ c) $m + 8 = 8$

2. (2 marks)

Solve: a) $4x = 20$ b) $x \div 3 = 6$ c) $7x = 56$

3. (2 marks)

Solve: a) $x - 9 = -2$ b) $x + 11 = 4$ c) $6x = -30$

4. (3 marks)

Solve: a) $x \div 5 = -4$ b) $-3x = 21$ c) $x + 2.5 = 6$

5. (3 marks)

Solve (letter on the right): a) $9 = x + 4$ b) $20 = 5x$ c) $-8 = x - 3$

6. (3 marks)

Form and solve: a) A number n times 4 is 28. b) 5 subtracted from p gives 12.

7. (3 marks)

Rectangle: width w , length $5w = 35$ cm. a) Equation in w . b) Find w . c) Perimeter.

8. (5 marks)

Each parcel costs $\pounds x$; 4 parcels cost $\pounds 26$. a) Equation. b) Price of one parcel. c) Most whole parcels for $\pounds 40$?